

# Scaling Portfolio-Level Reinforcement Learning via Anchor-Token Cross-Attention

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Architecture analysis and manuscript drafted with Claude (Anthropic)

September 2026

## Abstract

**TradingGame** is a rule-constrained portfolio-management reinforcement learning environment: an actor-critic policy chooses buy/sell/hold actions jointly across a candidate universe of  $N$  stocks under a shared cash constraint, trained with Proximal Policy Optimization (PPO). The current policy network aggregates cross-candidate context with full self-attention over all  $N$  candidates plus a portfolio token, an  $O(N^2)$  operation. This note (i) formalizes the environment and the existing architecture, (ii) reports a direct empirical characterization of that architecture’s GPU memory footprint — thirteen measurements on real hardware, fitted to a closed-form model with  $R^2 = 0.9999999$  — showing the design cannot reach a  $N=1,500$ -company universe at useful batch sizes, and (iii) proposes an  $O(N(K+2))$  replacement, *anchor-token cross-attention*, in which each candidate attends only to a small fixed set of  $K$  real anchor stocks plus separate macro and portfolio tokens, rather than to every other candidate or to a single pooled summary. We show the redesign is necessary but not sufficient: removing the quadratic term exposes a previously-dominated linear term, whose coefficient we also measure directly, giving a closed-form bound on the batch size the redesign alone can reach at a given  $N$ .

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## 1 Introduction

`TradingGame` trains a single reinforcement-learning policy to manage a simulated portfolio of Indian (NSE/BSE) equities under an explicit rule set: a fixed starting cash balance; a 0.5% fee on every transaction; sale proceeds that settle into spendable cash only after a two-trading-day delay; a one-trading-day lock-up before a newly-bought lot can be voluntarily sold again; and an unconditional forced exit after ten trading days. The policy observes a capped *candidate universe* of  $N$  stocks at each decision step and must choose, jointly, which to hold, sell, or buy — subject to a single shared cash pool, so the decisions are coupled across candidates, not independent per-stock classifications.

The current implementation (`ActorCriticPolicy` in `packages/TradingGame/src/policy.jl`) caps  $N$  at 60. Growing the candidate universe toward index scale ( $N \approx 1,500$ ) is desirable for realistic diversification, and is newly plausible now that training runs on a dedicated GPU rather than CPU alone. Section 4 shows why the current architecture cannot simply be pointed at a larger  $N$ : its cross-candidate attention mechanism is quadratic in  $N$ . Section 5 characterizes that cost directly, on real hardware, rather than by asymptotic argument alone. Section 6 derives a linear-cost replacement, and Section 7 shows why that replacement is a necessary but incomplete fix.

## 2 Related Work

**Self-attention and its cost.** Scaled dot-product self-attention [Vaswani et al., 2017] lets every element of a set condition on every other element, at  $O(n^2)$  time and memory in the set size  $n$ . This is the mechanism the current `ActorCriticPolicy` uses across candidates.

**Reducing attention to sub-quadratic cost.** Several strategies replace all-pairs attention with attention against a smaller, fixed reference set. Lee et al. [2019] introduce the *Induced Set Attention Block* (ISAB): rather than  $n$  elements attending to all  $n$  others, they attend to  $m \ll n$  learned *inducing points*, reducing cost from  $O(n^2)$  to  $O(nm)$  while remaining a permutation-invariant function of the input set — exactly the structure we adopt in Section 6, with the inducing points replaced by a small set of *real, interpretable* anchor stocks. Other lines of work achieve sub-quadratic attention via low-rank projection of the sequence dimension [Wang et al., 2020] or kernel approximations of the softmax [Choromanski et al., 2021]; we do not adopt these here because they trade the interpretability of "which stocks is candidate  $i$  attending to" for a compressed or randomized basis, which is a poor fit for a financial decision system where that interpretability is itself valuable.

**Recurrent per-element encoders.** Each candidate’s own price history is encoded independently with a Gated Recurrent Unit [Cho et al., 2014] before any cross-candidate mixing; this per-candidate encoding is where the dominant *linear-in- $N$*  cost identified in Section 7 originates, and it is unaffected by whichever cross-candidate attention pattern is used downstream.

**Combined policy/value networks trained by self-play or policy-gradient RL.** AlphaGo Zero and its generalization [Silver et al., 2017, 2018] share a trunk between a policy head and a value head, trained end-to-end — the same actor-critic head structure `ActorCriticPolicy` uses, though those systems operate over a fixed-size board rather than a variable, rule-constrained financial action space. We train with Proximal Policy Optimization [Schulman et al., 2017] and Generalized Advantage Estimation [Schulman et al., 2016], in the tradition of actor-critic methods such as A3C [Mnih et al., 2016]; optimization uses Adam [Kingma and Ba, 2015]. LeCun et al. [2015] give a general account of the representation learning this architecture relies on throughout.

**Internal precedent.** This repository’s own `StockSwingPredictor` package made the analogous choice once before, for a different model: its first architecture (`DualCNN_v1`) aggregated a target stock’s peer context by *mean-pooling* up to 150 peer embeddings; the second (`DualCNN_v2`) replaced that pooling with cross-attention, specifically because a flat average cannot express *which* peers matter for a given target. Section 6 applies the identical argument to `TradingGame`: an earlier draft of this proposal itself made the mean-pool mistake it is meant to avoid, by aggregating all  $N$  candidates into one pooled "market pulse" token before realizing this reintroduced exactly the selectivity loss that motivated `DualCNN_v2` in the first place.

### 3 Environment and Decision Problem

We formalize the rules enforced by `packages/TradingGame/src/env.jl` (`TradingGameRules.txt`, repo root) as a discrete-time Markov decision process.

**Portfolio state.** At trading-hour index  $t$ , the portfolio holds spendable cash  $C_t$ , a set of reserved-cash lots  $\mathcal{R}_t = \{(\rho_k, \tau_k)\}$  (amount, maturity trading-day index), and a set of open lots  $\mathcal{L}_t = \{(i_\ell, q_\ell, p_\ell, d_\ell)\}$  (stock index, quantity, entry price, entry trading-day index). Portfolio value is

$$V_t = C_t + \sum_{(\rho, \tau) \in \mathcal{R}_t} \rho + \sum_{(i, q, p, d) \in \mathcal{L}_t} q \cdot P_{i,t}, \quad (1)$$

where  $P_{i,t}$  is stock  $i$ ’s price at  $t$  — rules 2 and 6 of `TradingGameRules.txt`, restated identically.

**Settlement, lock-up, forced exit.** Let  $\delta(t)$  be the trading-day index of hour  $t$ . A reserved-cash lot  $(\rho, \tau)$  matures — moves into  $C_t$  — once  $\delta(t) \geq \tau$ , with  $\tau = \delta(t_{\text{sell}}) + \tau_{\text{settle}}$ ,  $\tau_{\text{settle}} = 2$  (rule 4). A lot  $(i, q, p, d)$  may be voluntarily sold only once  $\delta(t) - d \geq \tau_{\text{lock}}$ ,  $\tau_{\text{lock}} = 1$  (rule 9), and is *unconditionally* liquidated once  $\delta(t) - d \geq \tau_{\text{max}}$ ,  $\tau_{\text{max}} = 10$  (rule 8), independent of the policy’s action.

**Action space.** At each decision step the policy emits, jointly for every candidate  $i \in \{1, \dots, N\}$ , a discrete type and a continuous weight:

$$a_t = (a_{i,t}^{\text{type}}, w_{i,t})_{i=1}^N, \quad a_{i,t}^{\text{type}} \in \{\text{HOLD}, \text{SELL}, \text{BUY}\}, \quad w_{i,t} \in [0, 1]. \quad (2)$$

$w_{i,t}$  is only meaningful when  $a_{i,t}^{\text{type}} = \text{BUY}$ : it is the requested share of available cash, renormalized across every candidate selected BUY this step so that total notional-plus-fee never exceeds  $C_t$  (rule 5) — a deterministic masking layer (`resolve_actions` in `action.jl`), not something the policy must learn unaided. Every executed transaction pays a fee  $\varphi = 0.005$  of its notional value (rule 10), and a SELL is only ever offered to the policy for lots that have cleared the lock-up in Eq. (2)’s constraint.

**Reward.** Consistent with rule 11's "continuously maximize portfolio value,"

$$r_t = \log \frac{V_t}{V_{t-1}}, \quad (3)$$

computed every hourly bar (not only on decision bars), so unrealized mark-to-market movement is credited even when the policy takes no action. Because Eq. (1) already carries reserved cash at full value throughout its settlement delay, a sale's only effect on  $r_t$  is the fee  $\varphi$  — the reward requires no separate transaction-cost penalty term.

## 4 Baseline Architecture: Full Self-Attention

`ActorCriticPolicy` (`policy.jl`, current implementation) computes an embedding  $h_i \in \mathbb{R}^d$  for each candidate independently with a weight-shared GRU over its own  $T=120$ -bar,  $C=3$ -channel hourly window, fuses in per-candidate news and holding-state features, and forms one additional *portfolio/macro token*  $m$  from the macro time series and the global portfolio scalars. The  $N+1$  embeddings  $\{h_1, \dots, h_N, m\}$  are then passed through full multi-head self-attention [Vaswani et al., 2017]:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \quad Q = K = V = [h_1; \dots; h_N; m] \in \mathbb{R}^{(N+1) \times d}. \quad (4)$$

Every one of the  $N+1$  tokens is both a query and a key: the score matrix  $QK^\top$  has  $(N+1)^2$  entries, computed and differentiated once per attention head and once per batch element. This is the mechanism by which the network lets candidate  $i$ 's decision condition on candidate  $j$ 's current state — necessary for a *joint*, cash-constrained decision — but its cost scales as

$$\text{cost}_{\text{attn}}(N) = \Theta((N+1)^2). \quad (5)$$

Figure 1(a) shows this connectivity for a small illustrative  $N$ : every node connects to every other.

## 5 Empirical Characterization of the Baseline

Asymptotic complexity alone does not say where a design stops fitting a particular GPU, so Eq. (5) was measured directly rather than taken on faith. For each  $(N, B)$  pair, `ActorCriticPolicy` was constructed, moved to GPU, and run through one full forward pass, backward pass, and Adam optimizer step [Kingma and Ba, 2015] on random input of the correct shape; peak resident GPU memory was read from `CUDA.used_memory()` immediately after. All measurements are on a single NVIDIA GeForce RTX 5060 Ti (16.72 GB total,  $\approx 15.57$  GB usable after driver/OS reservation).

### 5.1 Fitted memory model

The thirteen successful measurements were fit by ordinary least squares to

$$\widehat{\text{mem}}(N, B) = c_0 + c_1 B + c_2 (NB) + c_3 (N^2 B) \quad [\text{GB}], \quad (6)$$

whose four terms correspond, respectively, to a fixed constant overhead, a per-batch-element cost independent of  $N$  (biases, small dense layers), a cost linear in both  $N$  and  $B$  (the GRU encoder

**Table 1:** Measured peak GPU memory, ActorCriticPolicy forward + backward + Adam step.

| $N$   | Batch $B$ | Peak memory (GB) | Result     |
|-------|-----------|------------------|------------|
| 60    | 4         | 0.122            | fits       |
| 60    | 8         | 0.250            | fits       |
| 60    | 16        | 0.493            | fits       |
| 60    | 32        | 0.986            | fits       |
| 60    | 64        | 1.971            | fits       |
| 60    | 128       | 3.941            | fits       |
| 60    | 256       | 7.882            | fits       |
| 500   | 4         | 1.083            | fits       |
| 500   | 8         | 2.165            | fits       |
| 500   | 16        | 4.329            | fits       |
| 500   | 32        | 8.657            | fits       |
| 500   | 64        | —                | <b>OOM</b> |
| 1,500 | 4         | 3.631            | fits       |
| 1,500 | 8         | 7.261            | fits       |
| 1,500 | 16        | —                | <b>OOM</b> |

and fusion/head layers, each processing  $N \times B$  independent sequences), and a cost quadratic in  $N$ , linear in  $B$  (self-attention, Eq. (5), batched). The fit is

$$c_0 = 9.710 \times 10^{-4}, \quad c_1 = 1.615 \times 10^{-5}, \quad (7)$$

$$c_2 = 5.090 \times 10^{-4}, \quad c_3 = 6.400 \times 10^{-8}, \quad (8)$$

with  $R^2 = 0.9999999$  across all thirteen points — residuals below  $4 \times 10^{-3}$  GB throughout, i.e. essentially exact given measurement granularity. This fit is the quantitative basis for every projection in the remainder of this note, and for the live calculator at [website/tradinggamescale.html](https://website/tradinggamescale.html).

## 5.2 A caveat: peak transient memory exceeds the fitted quantity

Eq. (6) predicts  $\widehat{\text{mem}}(1500, 16) \approx 14.5$  GB — comfortably under the 15.57 GB usable ceiling — yet  $N=1500, B=16$  was *measured* to fail with an out-of-memory error. `used_memory()` reports resident memory after the step completes; it understates the *peak* memory held transiently during backpropagation through attention, when the softmax output, its gradient, and several intermediate tensors coexist. We do not attempt to model that peak directly; instead, every capacity claim in this note treats the highest value that was *empirically observed to succeed* (8.657 GB, at  $N=500, B=32$ ) as the boundary of a verified-safe region, and treats predictions between that value and the raw usable ceiling as requiring direct verification before being relied on.

# 6 Proposed Architecture: Anchor-Token Cross-Attention

## 6.1 Design

Replace Eq. (4)’s all-pairs attention with attention against a small, fixed *context set*  $\mathcal{C}$ , built from three parts:

- **Anchor stocks.** A fixed subset  $\mathcal{A} \subset \{1, \dots, N\}$ ,  $|\mathcal{A}| = K$  (e.g. index or sector leaders). Their embeddings  $\{h_j : j \in \mathcal{A}\}$  are already computed as part of the per-candidate GRU pass over all  $N$  candidates — selecting them costs  $O(1)$ , with no additional encoding.
- **Macro token**  $m_{\text{macro}}$ , from the existing macro-series GRU, kept separate rather than pre-fused with portfolio state.
- **Portfolio token**  $m_{\text{port}}$ , from the global portfolio scalars, likewise kept separate.

$$\mathcal{C} = \{h_j : j \in \mathcal{A}\} \cup \{m_{\text{macro}}, m_{\text{port}}\}, \quad |\mathcal{C}| = K + 2. \quad (9)$$

Every candidate, including anchors themselves, queries against this fixed set:

$$z_i = \text{Attn}(Q=h_i, K=V=\mathcal{C}), \quad i = 1, \dots, N. \quad (10)$$

This is precisely the Induced Set Attention Block construction of Lee et al. [2019], with the inducing points replaced by *interpretable, real* anchor stocks and two designated context tokens, rather than learned abstract vectors — a deliberate choice for a system whose outputs should remain auditable (Section 2).

## 6.2 Why not a single pooled token

An intermediate design — reject in favor of Eq. (9) — would replace the anchor set with a single pooled summary,  $m_{\text{pool}} = \frac{1}{N} \sum_i h_i$ , and let every candidate attend only to  $\{m_{\text{pool}}, m_{\text{macro}}, m_{\text{port}}\}$ . This is cheaper still,  $O(N \cdot 3)$ , but every candidate then reads the *identical* pooled vector: attention over a one-element key set degenerates to a fixed (query-independent, up to normalization) linear combination, discarding exactly the selectivity that motivates attention over mean-pooling in the first place — the same reasoning **StockSwingPredictor** applied when it replaced **DualCNN\_v1**’s mean-pool with **DualCNN\_v2**’s cross-attention (Section 2). Real, distinguishable anchor stocks let different candidates learn different attention weights over  $\mathcal{A}$ .

## 6.3 Complexity

Attention cost is now

$$\text{cost}_{\text{attn}}(N, K) = \Theta(N \cdot (K + 2)), \quad (11)$$

linear in  $N$  for fixed  $K$ , versus Eq. (5)’s quadratic cost. Figure 1(b) shows the resulting connectivity: outer (candidate) nodes connect only to the small central context set, not to each other.

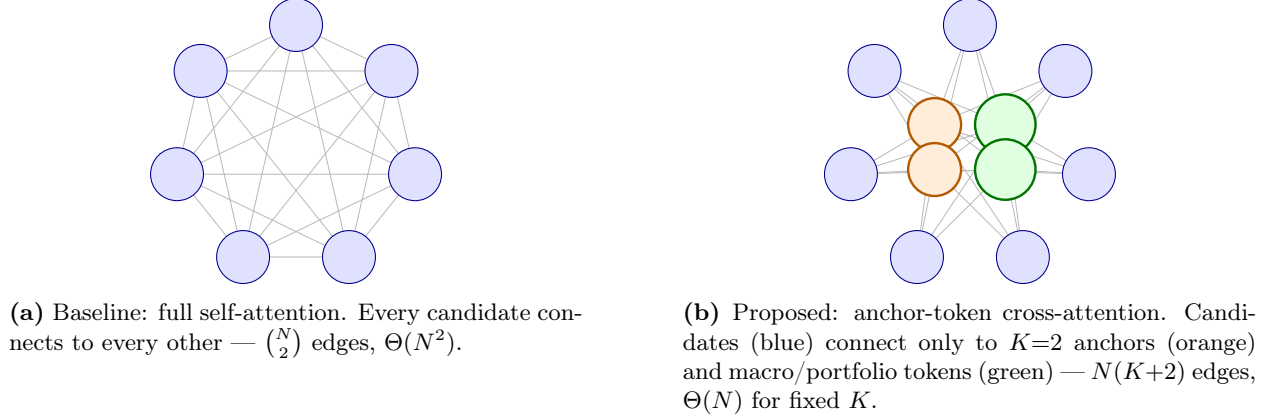
## 6.4 Pipeline

Figure 2 shows the full proposed forward pass. The per-candidate GRU encoder, and the actor/critic heads, are unchanged from the baseline; only the aggregation stage between them changes.

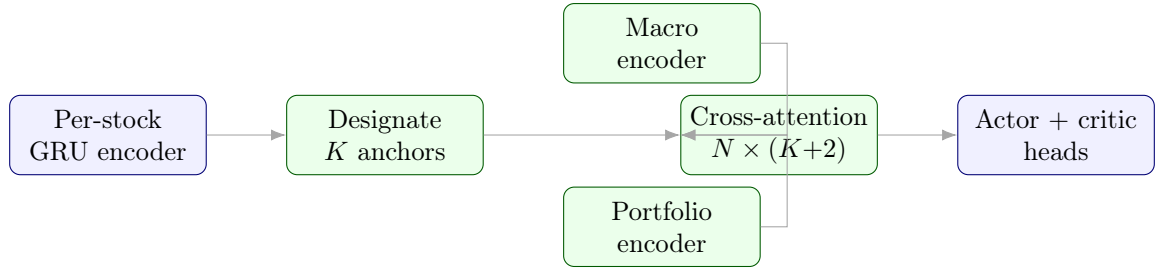
# 7 The Redesign Is Necessary but Not Sufficient

Substituting Eq. (11) for Eq. (5) inside the fitted model of Eq. (6) — reusing  $c_0, c_1, c_2$  unchanged, since the GRU/fusion/head costs they capture do not depend on the attention pattern, and reusing  $c_3$  as the measured per-key-value-pair cost of attention itself — gives a projected memory function for the proposed design:

$$\widehat{\text{mem}}_{\text{proposed}}(N, K, B) = c_0 + c_1 B + c_2 (NB) + c_3 (N(K + 2)B). \quad (12)$$



**Figure 1:** Attention connectivity, illustrated at a small  $N$  for legibility.



**Figure 2:** Proposed policy pipeline. Blue: unchanged from the baseline. Green: new or restructured relative to Section 4.

As a consistency check, setting  $K = N - 2$  (a degenerate anchor set equal to the whole candidate universe) recovers Eq. (6) exactly, as it must:  $\widehat{\text{mem}}_{\text{proposed}}(N, N-2, B) = \widehat{\text{mem}}(N, B)$ .

Table 2 evaluates both models at  $N=1500$ ,  $K=8$ . The proposed design does reduce memory — by roughly 16% at  $B=32$  — but *both* figures exceed this GPU’s usable ceiling. The reason is visible directly in Eq. (12): at  $N=1500$ , the linear term  $c_2(NB)$  already dominates the total almost completely ( $c_2 \cdot 1500 \cdot 32 \approx 24.4$  GB, versus a quadratic-term contribution, at  $K=8$ , of only  $c_3 \cdot 1500 \cdot 10 \cdot 32 \approx 0.031$  GB). Eq. (11)’s asymptotic improvement is real, but at  $N=1500$  it has already stopped being the binding constraint.

**Table 2:** Baseline vs. proposed, projected at  $N=1500$ ,  $K=8$ .

|           | Baseline (GB) | Proposed, $K=8$ (GB) |
|-----------|---------------|----------------------|
| $B = 4$   | 3.63          | 3.06                 |
| $B = 8$   | 7.26          | 6.12                 |
| $B = 32$  | 29.04         | 24.46                |
| $B = 256$ | 232.3         | 195.7                |

Solving Eq. (12) for the largest batch size under a memory budget  $M$ ,

$$B_{\max}(N, K, M) = \frac{M - c_0}{c_1 + c_2 N + c_3 N(K + 2)}, \quad (13)$$

gives, at  $N=1500$ ,  $K=8$ :  $B_{\max} \approx 11.4$  under the verified-safe budget ( $M=8.657$  GB) and  $B_{\max} \approx$

20.4 under the raw usable ceiling ( $M=15.57$  GB, subject to the Section 5.2 caveat). Neither approaches the batch sizes (128–256) that fit easily at  $N=60$  today. Reaching them at  $N=1500$  requires a second lever independent of the attention redesign — most plausibly a smaller per-candidate embedding width, mixed-precision training, or gradient checkpointing through the GRU — which we leave as future work (Section 9).

## 8 Training

The policy is trained with Proximal Policy Optimization [Schulman et al., 2017] using Generalized Advantage Estimation [Schulman et al., 2016]; the loop is unaffected by the architecture change proposed here beyond the forward-pass cost analyzed above. Algorithm 1 summarizes the current implementation (`ppo.jl`, `train.jl`).

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### Algorithm 1 TradingGame training iteration

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- 1: Roll out one full episode with the current policy, sampling  $a_{i,t}^{\text{type}} \sim \text{Categorical}(\text{softmax}(\ell_{i,t}))$  per candidate; record  $(s_t, a_t, \log \pi_{\text{old}}(a_t|s_t), r_t, V(s_t))_{t=1}^T$
  - 2: Compute advantages via GAE:  $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ ,  $\hat{A}_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}$ ; normalize  $\hat{A}_t$
  - 3: **for**  $K_{\text{epochs}}$  passes over shuffled minibatches **do**
  - 4:    $\rho_t(\theta) \leftarrow \exp(\log \pi_{\theta}(a_t|s_t) - \log \pi_{\text{old}}(a_t|s_t))$
  - 5:    $\mathcal{L}^{\text{CLIP}} \leftarrow \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)]$
  - 6:    $\mathcal{L} \leftarrow -\mathcal{L}^{\text{CLIP}} + c_v \mathbb{E}_t[(V_{\theta}(s_t) - \hat{R}_t)^2] - c_e \mathbb{E}_t[\mathcal{H}(\pi_{\theta}(\cdot|s_t))]$
  - 7:   Update  $\theta$  with Adam [Kingma and Ba, 2015]
  - 8: **end for**
  - 9: Checkpoint if the (held-out, greedy) episode return improved
- 

The continuous buy-weight  $w_{i,t}$  (Eq. (2)) is used deterministically —  $w_{i,t} = \sigma(\text{logit})$ , renormalized by the masking layer — rather than sampled from a second distribution; PPO’s log-probability ratio  $\rho_t$  is computed over the categorical  $a^{\text{type}}$  choice only. This keeps the stochastic side of a hybrid discrete/continuous action space tractable for a hand-rolled implementation, at the cost of not learning position-sizing stochastically. None of this changes under the proposed architecture, which affects only how  $s_t$  is encoded into  $\ell_{i,t}$  and  $V(s_t)$ .

## 9 Discussion and Open Questions

- **Anchor selection.** Candidates include index constituents (e.g. Nifty 50), the top- $K$  candidates by market capitalization, or one representative per major sector. Any choice must be a subset of the episode’s own candidate universe, so no extra encoding pass is needed.
- **Anchor stability.** Fixing  $\mathcal{A}$  per episode keeps  $K$  genuinely constant for Eq. (11)’s accounting; re-selecting anchors more frequently (e.g. today’s top movers) could be more informative but reopens a small  $O(N)$  selection cost per step and makes the attended set non-stationary within an episode.
- **The linear bottleneck (Section 7)** is, on the evidence here, the harder remaining problem: it is intrinsic to encoding  $N$  independent price histories, not to any particular cross-candidate mixing strategy.

- **GPU wiring.** `ppo.jl/train.jl` currently run entirely on CPU (a deliberate scoping decision for the initial PPO implementation); every measurement in this note characterizes the network in isolation, not the full rollout-and-update loop.
- **assemble\_observation is a separate, CPU-bound cost** — a per-candidate Julia loop run once per environment step — outside the scope of the GPU analysis here and worth profiling independently as  $N$  grows.

## 10 Conclusion

Full self-attention across the candidate universe, as implemented today, is quadratic in  $N$  and — measured directly, not merely argued asymptotically — exhausts a 16.7GB GPU at  $N=1500$  between batch sizes 8 and 16. Anchor-token cross-attention, restricted to a small fixed context set of real anchor stocks plus separate macro and portfolio tokens, is linear in  $N$  and recovers exactly the selectivity a pooled-summary alternative would have discarded. It reduces memory meaningfully (Table 2) but does not, by itself, make  $N=1500$  training cheap: a linear per-candidate encoding cost, fit from the same measurements (Eq. (6)), remains and now dominates. Both findings — the quadratic bottleneck and its linear successor — come from the same thirteen-point empirical characterization and the same closed-form model, which is the artifact we consider most durable here: not a specific proposed architecture, but a validated method for answering "does this fit on this GPU" without guessing.

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